Block Term Decomposition-Guided Frequency Mamba Modulation for Hyperspectral Image Fusion

Yan Li, Chuangjie Fang, Zhijie Chen, Yiqun Meng, Honghui Xu, *Member, IEEE*, Hao Wang, and Yi-Peng Liu, *Member, IEEE*

***Abstract*—Owing to the complementary spatial–spectral characteristics of hyperspectral images (HSIs) and multispectral images (MSIs), HSI–MSI fusion has become an effective approach for generating high-resolution HSIs (HR-HSIs). Recent deep learning (DL) frameworks, particularly Mamba-based architectures, provide efficient long-range dependence modeling with near-linear complexity. Nevertheless, they often overlook the intrinsic low-rank structure of hyperspectral data and are largely insensitive to frequency-domain information. To address these limitations, we propose block term decomposition-guided frequency Mamba modulation (BFMM), a low-rank tensor modulation framework built upon block term decomposition (BTD). BFMM represents features as a sum of multiple Tucker components, each reconstructed by the multimode product of a learnable Tucker core and low-rank factor matrices, thereby preserving structured spatial–spectral dependencies. A frequency Mamba joint modulation (FMJM) unit learns frequency-aware low-rank Tucker cores through adaptive frequency conditioning, and a Kronecker-rank interaction modulator (KRIM) learns spatial–spectral factor matrices via separable channel-rank guidance in the form of Kronecker products. The modulated core tensors are sequentially contracted with the learned factors through mode-*n* products, thereby reconstructing low-rank feature tensors whose aggregation yields structurally interpretable and information-dense representations. Across three public datasets, BFMM achieves state-of-the-art reconstruction performance with a favorable accuracy–complexity tradeoff, and is further validated on the WHU-MHF and WV-3 datasets. The source code is available at https://github.com/lirui818/BFMM**

***Index Terms*—Block term decomposition (BTD), frequency-domain modulation, hyperspectral image (HSI) fusion, low-rank modeling, visual state-space modeling.**

I. Introduction

HYPERSPECTRAL images (HSIs) provide rich spectral information that enables accurate characterization of the material composition of ground objects, and are widely used in tasks such as classification [1], [2], segmentation [3], [4], and detection [5], [6]. However, owing to various hardware limitations [7], an inherent tradeoff between spectral resolution and spatial resolution exists. Increasing the spatial resolution by redesigning hyperspectral imaging systems is both technically challenging and cost-prohibitive. As an alternative, fusing low-resolution HSIs (LR-HSIs) with high-resolution multispectral images (HR-MSIs) to generate high-resolution HSIs (HR-HSIs) has emerged as a promising and cost-effective strategy for acquiring high-quality hyperspectral data.

Existing methods can be broadly categorized into two groups: model-based schemes and deep-learning (DL)-based approaches. Model-based techniques include component substitution (CS) [8], multiresolution analysis (MRA) [9], and variational optimization (VO) [10]. These methods impose explicit mathematical priors or physical constraints to guide the reconstruction process [11]. However, hand-crafted or subjective priors often fail to adapt to nonlinear degradation, complex noise, and large-scale high-dimensional data, thus leading to suboptimal performance. In addition, many of these methods rely on iterative solvers that are computationally demanding and highly sensitive to hyperparameters, thereby limiting their robustness and scalability in real-world applications.

DL [12], [13], [14] has propelled rapid progress in HSI super-resolution (HSI-SR) through powerful nonlinear representations and end-to-end training. Convolutional models exploit translation equivariance and weight sharing for efficient local feature extraction, yet fixed receptive fields limit the modeling of long-range dependencies. Transformer-based methods [15], [16], [17] capture global dependencies via self-attention, thus partially addressing the limitations of convolutional neural networks (CNNs) [12], [18]. However, their quadratic complexity with respect to the input length results in prohibitively high computational costs for high-resolution scenarios. The recently proposed Mamba [19] architecture introduces a state-space formulation with selective scanning, which enables linear-time and linear-memory modeling of long-range interactions, while supporting streaming inference and stable spatiotemporal recursion, properties that make it particularly appealing for long sequences and wide fields of view. However, practical HSI-SR on satellite data often suffers from the loss of perceptually critical frequencies, as vanilla Mamba processes spatial tokens with near-uniform weighting and is largely frequency-insensitive, which limits its ability

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to adaptively amplify frequency-informative cues. Overall, the current pipelines fail to fully exploit frequency-aware modeling and structural priors that can regularize high-dimensional features.

In this context, hyperspectral data exhibit strong interband correlation and high spatial regularity, and deep features in HSI-SR are, therefore, typically low-rank and redundant [20]. Such priors are effective for capturing the intrinsic organization of high-dimensional features [21], [22], yet many existing deep models do not fully exploit them, which limits both accuracy and robustness. To leverage the low-rank structure, LTRN [23] combines canonical polyadic (CP) decomposition with neural networks and injects CP priors through learned rank-1 components, thereby reducing redundancy. However, strictly separable rank-1 atoms represent features as linear combinations of outer products, which attenuates higher-order spatial and spectral dependencies and can cause the loss of informative low-rank variations or spatial and spectral aliasing. Real scenes are strongly anisotropic across the height, width, and channel dimensions, so a single isotropic rank cannot adapt to diverse structural complexities without sacrificing fidelity or efficiency. In contrast, block term decomposition (BTD) factorizes a tensor into a sum of low-rank Tucker blocks, striking a balance between expressive power and parameter efficiency and providing a more flexible way to encode anisotropic structures.

To address these challenges, we propose block term decomposition-guided frequency Mamba modulation (BFMM), a low-rank tensor modulation network built upon BTD. The model factorizes input features into a compact core tensor and mode-specific factor matrices across the height, width, and spectral dimensions, thereby preserving cross-modal dependencies while enabling anisotropic rank allocation. In the core space, we further introduce a frequency Mamba joint modulation (FMJM) module. Its frequency-compensation branch guides Mamba-based core modulation to compensate for the frequency-domain degradation commonly observed in satellite imaging and to enhance discriminative high-frequency details. Meanwhile, the Mamba pathway captures long-range spatial context with near-linear complexity, making the framework well-suited to high-resolution remote sensing scenarios. Complementing this core pathway, a Kronecker-rank interaction modulator (KRIM) learns data-adaptive factor matrices through a separable interaction between channel-importance vectors and rank-direction weights. This design effectively mitigates the information flattening typical of naive pooling and enhances overall robustness. The modulated Tucker core is then contracted with the factor matrices through sequential modewise products to reconstruct multiple low-rank tensor components. These components are then aggregated according to the additive structure of BTD to facilitate the efficient integration of high-dimensional features and robust low-rank approximation. By unifying frequency-aware modulation, state-space modeling, and low-rank tensor representation in an end-to-end trainable architecture, BFMM embeds interpretable structural priors while maintaining efficient

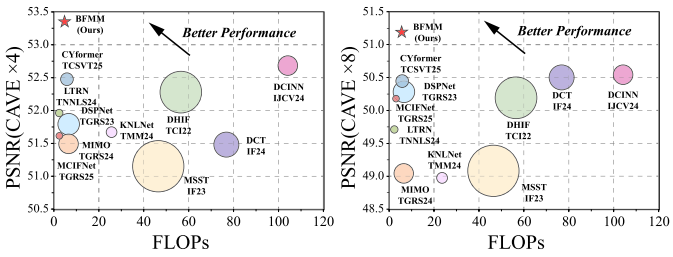


Fig. 1. Comparison of our method with competing methods on the CAVE dataset under the ×4 and ×8 settings. Methods closer to the top-left corner achieve higher PSNR with lower computational cost. Bubble size indicates the number of parameters.

global modeling capacity. The main contributions of BFMM are summarized as follows.

1) We develop an HSI-SR framework in which BTD is integrated into a deep model to introduce explicit low-rank structural constraints across spatial and spectral modes.
2) We propose an FMJM module that estimates the Tucker core by integrating adaptive frequency compensation with linear-complexity long-range spatial modeling via a visual state-space model (SSM), yielding stable and information-rich core representations.
3) We develop a KRIM that generates data-adaptive factor matrices through Kronecker-product-based channel-rank interactions, enabling parameter-efficient spatial–spectral factor modulation while preserving first-order spatial statistics.
4) Experiments on three public datasets show that the proposed method generates high-fidelity HR-HSIs with a favorable accuracy–complexity tradeoff, as illustrated in Fig. 1, while additional real-world evaluations validate its practical applicability and cross-task generalization.

II. Related Work

A. HSI Fusion

Model-based and DL methods constitute the two principal lines of research in spatial–spectral fusion. Conventional model-based approaches [24], [25], [26] exploit explicit priors that are grounded in well-defined mathematical formulations or physical principles, thus ensuring interpretability. However, their reliance on linear low-rank assumptions renders them ill-suited for nonlinear mixing, interspectral interactions, and cross-modal relationships, which are common in hyperspectral data. In addition, iterative optimization pipelines scale poorly because the computational burden increases rapidly with tensor order [27], [28], and they are often sensitive to hyperparameters, which hinders convergence and large-scale deployment.

The advent of deep CNNs [29] has catalyzed rapid progress in HSI fusion. To leverage the full representational capacity of CNNs, several innovative architectures have been developed. Hu et al. [30] incorporated spatial and channel attention mechanisms to better preserve spatial details. Similarly, Ran et al. [18] introduced a CNN-based fusion

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framework that employs recursive and progressive strategies for high-resolution guidance. Furthermore, Li et al. [31] designed an X-shaped interactive autoencoder, featuring cross-modality mutual learning to extract informative spatiospectral features [32]. Dian et al. [33] proposed a model-guided cross-fusion network for spectral super-resolution, which embeds an explicit imaging model to guide cross-modal feature fusion. To mitigate the reliance on large-scale paired training data, Dian et al. [34] further proposed a zero-shot hyperspectral sharpening strategy that performs internal learning at test time for each observed pair. Nonetheless, CNN-based models emphasize local perception through translation-equivariant filters and efficient weight sharing, while the fixed receptive fields of such models tend to limit long-range context aggregation.

As transformer-based methods [35], [36], [37] have gained prominence in computer vision, they have been widely applied to HSI-SR and have shown notable performance. Hu et al. [38] proposed the first transformer-based architecture for HSI–MSI fusion to globally exploit spectral information, and subsequent attention-based designs have further advanced the state of the art. Nevertheless, the quadratic complexity of self-attention with respect to input length poses challenges for deployment in high-resolution and long-sequence settings.

B. State-Space Models

SSMs have recently been explored as efficient alternatives to self-attention for long-range dependence modeling. Mamba [19] further improves the scalability of SSMs, enabling long-range context aggregation with linear complexity. This property is particularly attractive for high-resolution remote sensing imagery, where the quadratic cost of self-attention can become computationally prohibitive. Motivated by this advantage, Mamba has been introduced into the image domain [39], [40]. For example, VMamba [41] develops a 2-D selective scan to preserve spatial continuity while maintaining global receptive fields. Recent Mamba-based fusion methods [42], [43] incorporate SSM blocks to capture spatial dependencies and facilitate cross-modal information integration with reduced computational cost. However, these designs primarily rely on spatial SSM (Spatial SSM) modules and do not explicitly address the frequency degradation commonly observed in remote sensing imaging. To address this limitation, our method uses Mamba to guide frequency-aware Tucker-core modulation within a low-rank tensor formulation. The Tucker core is refined under joint spatial–frequency conditioning while retaining the linear-complexity global aggregation capability of Spatial SSM modules. This design helps recover attenuated frequency cues, stabilizes rank allocation during Tucker projection, and yields a more informative representation than purely spatial modeling.

C. Frequency-Based Deep Models

The Fourier transform provides a compact representation of global frequency content, offering a powerful complement to local spatial cues in image fusion [44]. Recent methods have increasingly sought to couple spatial features with frequency-domain priors. FISCNet [45] integrates frequency-domain phase information with spatial features to better preserve salient structures, while SFDFusion [46] adopts a dual-branch design to refine spatial details and improve efficiency via FFT-based processing. HFIN [47] further decomposes inputs into global and local Fourier components to strengthen spatial–frequency relations. In addition to directly introducing frequency-domain priors into fusion networks, recent work has also explored frequency cues as feature-level representation enhancers. Xiao et al. [48] introduced adaptive frequency selection within a Mamba-based pipeline to complement SSM long-range modeling and emphasize informative components for image-level reconstruction. Their work demonstrates the value of frequency cues for spatial detail recovery in large-scale remote sensing images. However, for HSI-SR, frequency compensation should be coordinated with spectral-correlation-aware representation learning [49]. To this end, the proposed frequency-aware design learns complementary frequency responses within Mamba-guided Tucker-core estimation, providing stable frequency cues for long-range spatial aggregation and subsequent modewise spatial–spectral reconstruction.

III. Methodology

A. Preliminaries

We consider two images captured from the same scene, where one image is an HR-MSI $\mathcal{Z} \in \mathbb{R}^{H\times W\times s}$, which provides higher spatial resolution but limited spectral details, and the other image is an LR-HSI $\mathcal{Y} \in \mathbb{R}^{h\times w\times S}$, which contains rich spectral information but has limited spatial resolution. The core objective of HSI-SR is to reconstruct an HR-HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$ that preserves fine spatial textures while maintaining spectral consistency. In this process, the height, width, and number of spectral bands of the images are denoted by H and $h(H > h)$, W and $w(W > w)$, and S and $s(S > s)$, respectively.

B. Overall Architecture

The overall architecture of the proposed BFMM is illustrated in Fig. 2. Given the LR-HSI $\mathcal{Y}$ and HR-MSI $\mathcal{Z}$, we first upsample $\mathcal{Y}$ via bicubic interpolation to obtain $\mathcal{Y}_{\text{up}} \in \mathbb{R}^{H\times W\times S}$, whose spatial dimensions match those of $\mathcal{Z}$. The two inputs are concatenated and passed through a 3×3 convolution to extract shallow features $\mathcal{X}_0 \in \mathbb{R}^{H\times W\times C}$, which is expressed as

$$\begin{aligned} \mathcal{Y}_{\text{up}} &= \text{Up}(\mathcal{Y}) \\ \mathcal{X}_0 &= \text{Conv}_{3\times 3}\left(\left[\mathcal{Y}_{\text{up}},\ \mathcal{Z}\right]\right) \in \mathbb{R}^{H\times W\times C} \end{aligned} \tag{1}$$

where $\text{Up}(\cdot)$ denotes bicubic upsampling, $\text{Conv}_{3\times 3}(\cdot)$ represents a 3×3 convolution, and C denotes the spectral feature dimension.

The extracted features are subsequently processed through T successive low-rank reconstruction blocks (LRRBs), each containing a BTD residual (BTDR) module and a multiscale

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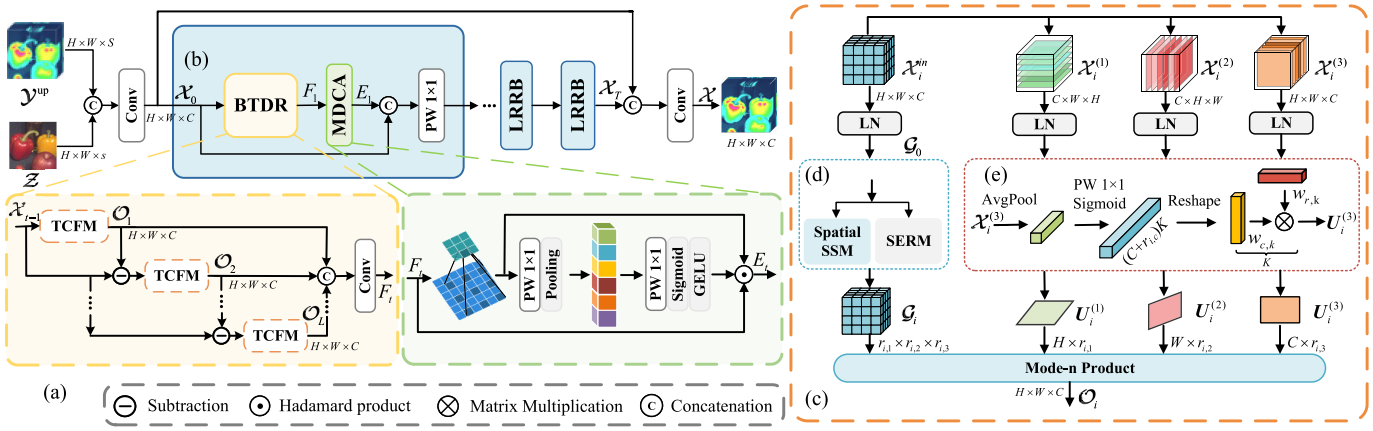


Fig. 2. Overall architecture of the proposed BFMM framework. (a) BFMM takes the LR-HSI and HR-MSI as inputs and progressively fuses them to reconstruct the HR-HSI. (b) Cascade of LRRBs refines the fused features, where each LRRB consists of a BTDR unit for structured low-rank feature factorization and an MDCA unit for lightweight channel recalibration. (c) TCFM serves as the core tucker modeling unit by integrating (d) FMJM for spatial–frequency core modulation and (e) KRIM for generating data-adaptive factor matrices, followed by successive mode-n products for low-rank tensor reconstruction.

dilated channel attention (MDCA) module, to achieve deep spatial–spectral integration. The update at stage t is

$$\begin{aligned} \boldsymbol{F}_t &= \mathrm{BTDR}\left(\boldsymbol{\mathcal{X}}_{t-1}\right) \\ \boldsymbol{E}_t &= \mathrm{MDCA}\left(\boldsymbol{F}_t\right) \\ \boldsymbol{\mathcal{X}}_t &= \mathrm{PWConv}\left(\left[\boldsymbol{F}_t,\ \boldsymbol{E}_t\right]\right), \quad t = 1, \ldots, T \end{aligned} \tag{2}$$

where $\boldsymbol{F}_t$ denotes the low-rank component that is produced by the BTDR at stage t, $\boldsymbol{E}_t$ denotes the MDCA-refined feature, and PWConv$(\cdot)$ denotes a 1×1 pointwise convolution.

The BTDR module performs residual BTD by cascading multiple Tucker core factor modulation (TCFM) units on a subtractive residual path. In each unit, the core tensor is refined within the frequency-aware state space, while the factor matrices are learned in a channel-rank manner. This process yields a complementary low-rank component, which is subtracted from the running residual. This design encourages nonredundant three-mode modeling across height, width, and spectrum, thereby enhancing spatial detail while preserving spectral fidelity. The MDCA branch captures local cross-channel interactions and adaptively recalibrates channel responses at minimal cost. The outputs from the BTDR and MDCA modules are concatenated along the channel dimension and projected by a pointwise convolution to update the stage. Finally, after T stages, the initial shallow feature and the final deep feature are concatenated and mapped by a 3×3 convolution to produce the final estimate

$$\boldsymbol{\mathcal{X}} = \mathrm{Conv}_{3\times 3}\left(\mathrm{Concat}\left(\boldsymbol{\mathcal{X}}_0, \boldsymbol{\mathcal{X}}_T\right)\right) \tag{3}$$

where Concat$(\cdot)$ denotes concatenation along the feature-channel dimension.

C. BTD Residual

Tensor decomposition reveals latent relationships within multidimensional data by representing the original tensor through lower-dimensional factor tensors. In this work, we design the BTDR module on the basis of the principle of BTD, which effectively captures low-rank structures in hyperspectral

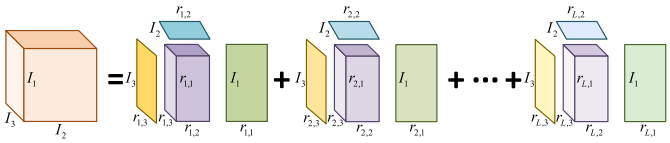


Fig. 3. Visual display of BTD.

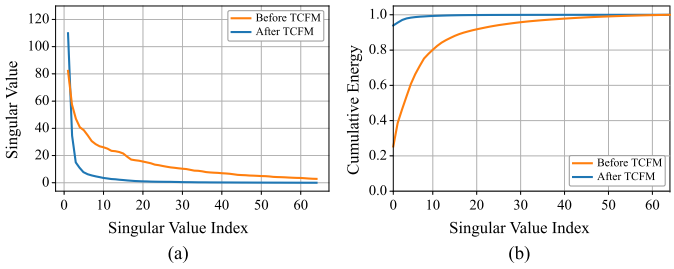


Fig. 4. Comparison of the spectral properties of the feature representations before and after the TCFM module. (a) Singular value spectra, where the horizontal axis denotes the singular value index and the vertical axis shows the singular value magnitude. (b) Cumulative energy curves of the singular values, reporting the fraction of total energy captured by the first k singular values.

data. Before introducing the BTDR module, we first outline the key concepts that underlie BTD.

BTD represents a tensor as a sum of multiple low-rank Tucker blocks, each of which comprises its own core tensor and a set of mode-specific factor matrices, as shown in Fig. 3. This structure enables factorized modeling of intricate patterns across modes. By decomposing the input into several structured components, BTD effectively captures diverse spatial and structural patterns within multiway data. Formally, for an N-order tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{I_1 \times I_2 \times, \ldots, \times I_N}$, the BTD formulation is given by

$$\boldsymbol{\mathcal{X}} = \sum_{i=1}^{L} \boldsymbol{\mathcal{G}}_i \times_1 \boldsymbol{U}_i^{(1)} \times_2 \boldsymbol{U}_i^{(2)}, \ldots, \times_N \boldsymbol{U}_i^{(N)} \tag{4}$$

where L represents the number of block terms. For each block term $i \in \{1, \ldots, L\}$, $\boldsymbol{\mathcal{G}}_i \in \mathbb{R}^{r_{i,1} \times r_{i,2} \times, \ldots, \times r_{i,N}}$ represents the core

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tensor of the ith block term. $\boldsymbol{U}_i^{(n)} \in \mathbb{R}^{I_n \times r_{i,n}}$ is its mode-n factor matrix with $n \in \{1, \ldots, N\}$, and the operator $\times_n$ denotes the mode-n product. The tuple $\{r_{i,1} \times r_{i,2} \times , \ldots, \times r_{i,N}\}$ defines the Tucker rank of the ith block term and determines the dimensionality reduction in each mode.

Unlike CP, which restricts every block to rank 1 across all modes, BTD retains a low-rank structure while modeling richer intrablock dependencies and multimodal interactions. This improves expressiveness and robustness, and often requires fewer blocks and offers better resilience to noise and collinearity. On the basis of the above theoretical foundation, we design the BTDR module to instantiate a BTD-inspired decomposition process by representing features as sums of multiple TCFM blocks. Each TCFM block adaptively learns anisotropic factor matrices and a modulated core tensor to capture mode-specific spatial and spectral structures. This design mimics the low-rank modeling nature of BTD, thus increasing representational capacity while maintaining computational efficiency.

Building on this BTD-inspired perspective, we instantiate the above idea in network design through the BTDR module. As shown in Fig. 2(c), we construct the BTDR module by cascading L TCFM blocks in a residual manner. The input tensor $\boldsymbol{\mathcal{X}}_{t-1}$ of the tth BTDR is first processed by the initial TCFM block to produce $\boldsymbol{\mathcal{O}}_1$. The second block then learns $\boldsymbol{\mathcal{O}}_2$ from the residual $\boldsymbol{\mathcal{X}}_{t-1} - \boldsymbol{\mathcal{O}}_1$, and subsequent blocks continue this process by refining the representation through features that were not captured in the earlier stages. Consequently, the output of the ith TCFM, $\boldsymbol{\mathcal{O}}_i$, can be expressed as

$$\boldsymbol{\mathcal{O}}_i = \mathrm{TCFM}_i\left(\boldsymbol{\mathcal{X}}_{t-1} - \sum_{j=1}^{i-1} \boldsymbol{\mathcal{O}}_j\right). \tag{5}$$

Each TCFM block learns a low-rank Tucker representation to effectively model complex high-order interactions across spatial and spectral dimensions. This progressive residual structure enables each TCFM block to capture complementary low-rank components within the same spatial region of a heterogeneous hyperspectral scene, thereby reducing redundancy, mitigating information loss, and facilitating more comprehensive feature modeling.

After all L TCFM blocks generate their respective low-rank Tucker representations, the resulting outputs are concatenated along the channel dimension and subsequently fused via a 3×3 convolution to produce the final output. This convolutional fusion adaptively integrates complementary frequency-aware features that are captured at different residual stages, thereby increasing the model's representational flexibility and expressive capacity. Formally, the final output of the tth BTDR, $\boldsymbol{F}_t$, can be expressed as

$$\boldsymbol{F}_t = \mathrm{Conv}_{3\times 3}\left(\mathrm{Concat}\left\{\boldsymbol{\mathcal{O}}_i \,\middle|\, i = 1, \ldots, L\right\}\right) \tag{6}$$

where Concat$(\cdot)$ denotes channelwise concatenation.

D. Tucker Core Factor Modulation

As illustrated in Fig. 2(c), we introduce the TCFM module to jointly estimate a compact core tensor and the corresponding mode-specific factor matrices, thereby constructing

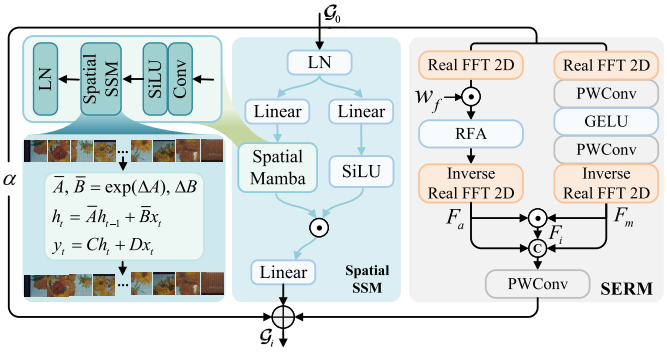


Fig. 5. Overview of the detailed structure of the proposed FMJM module.

a Tucker-structured low-rank tensor representation. To examine its effect on feature compactness, Fig. 4 reports the singular value spectra and cumulative energy curves of the feature tensor before and after TCFM. The representation after TCFM exhibits a faster singular value decay in Fig. 4(a) and reaches energy saturation with fewer dominant components in Fig. 4(b), indicating that TCFM produces a more compact low-rank representation. Within the BTDR framework, each TCFM unit corresponds to one Tucker block in the BTD formulation. Following the principle of Tucker decomposition, TCFM jointly models the core tensor and mode-specific factors to capture high-order spatial–spectral interactions. Unlike conventional iterative decomposition methods, TCFM directly estimates the core tensor and factor matrices through a deep neural network, which improves modeling efficiency and enables seamless integration into end-to-end optimization.

To obtain a compact and expressive core tensor representation within each TCFM module, the residual input $\boldsymbol{\mathcal{X}}_i^{\mathrm{in}}$ is first normalized and denoted as the precore feature $\boldsymbol{\mathcal{G}}_0 \in \mathbb{R}^{H\times W\times C}$. The FMJM module performs spatial–frequency modulation on $\boldsymbol{\mathcal{G}}_0$ to estimate the Tucker core $\boldsymbol{\mathcal{G}}_i \in \mathbb{R}^{r_{i,1}\times r_{i,2}\times r_{i,3}}$. The core estimation process is formulated as

$$\boldsymbol{\mathcal{G}}_i = \mathrm{FMJM}\left(\boldsymbol{\mathcal{G}}_0\right). \tag{7}$$

The FMJM unit consists of two complementary components: a spectral energy reallocation modulator (SERM) and a Spatial SSM module. SERM provides frequency-aware compensation for Tucker-core modulation by combining radial-frequency activation, learnable complex channel calibration, nonlinear frequency mixing, and interaction-based response coupling. In parallel, the Spatial SSM module captures long-range spatial dependencies with linear complexity, enabling efficient context aggregation for high-resolution inputs. By jointly modeling frequency-enhanced responses and spatial context, FMJM adaptively guides Tucker-core estimation and modewise low-rank reconstruction, thereby mitigating structural distortion and maintaining cross-domain consistency.

In parallel to core modulation, we derive mode-specific factor matrices that parameterize the Tucker subspaces. Linear projections along the height, width, and channel modes produce three dimension-specific representations

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$\boldsymbol{\mathcal{X}}_i^{(1)} \in \mathbb{R}^{C\times W\times H}$, $\boldsymbol{\mathcal{X}}_i^{(2)} \in \mathbb{R}^{C\times H\times W}$, and $\boldsymbol{\mathcal{X}}_i^{(3)} \in \mathbb{R}^{H\times W\times C}$, respectively. On this basis, we propose a KRIM that adaptively learns mode-specific factor matrices $\boldsymbol{U}_i^{(1)} \in \mathbb{R}^{H\times r_{i,1}}$, $\boldsymbol{U}_i^{(2)} \in \mathbb{R}^{W\times r_{i,2}}$, and $\boldsymbol{U}_i^{(3)} \in \mathbb{R}^{C\times r_{i,3}}$ through a separable interaction mechanism between channel-importance vectors and rank-direction weights, where the values $r_{i,1}$, $r_{i,2}$, and $r_{i,3}$ represent the Tucker ranks for the ith block term along the height, width, and channel modes, respectively. Accordingly, the three factor matrices that correspond to the different modes can be expressed as

$$\begin{aligned}\boldsymbol{U}_i^{(1)} &= \mathrm{KRIM}\left(\mathrm{LN}\left(\boldsymbol{\mathcal{X}}_i^{(1)}\right)\right)\\ \boldsymbol{U}_i^{(2)} &= \mathrm{KRIM}\left(\mathrm{LN}\left(\boldsymbol{\mathcal{X}}_i^{(2)}\right)\right)\\ \boldsymbol{U}_i^{(3)} &= \mathrm{KRIM}\left(\mathrm{LN}\left(\boldsymbol{\mathcal{X}}_i^{(3)}\right)\right).\end{aligned} \tag{8}$$

This design mitigates the information flattening that is typically induced by naive pooling operations, while improving both robustness and interpretability in low-rank representation learning. The refined Tucker core is subsequently reprojected through successive mode-n products with its factor matrices, thereby enabling coherent feature aggregation and a stable, structure-preserving low-rank approximation in high-dimensional space. The complete reconstruction process can be formulated as

$$\boldsymbol{\mathcal{O}}_i = \boldsymbol{\mathcal{G}}_i \times_1 \boldsymbol{U}_i^{(1)} \times_2 \boldsymbol{U}_i^{(2)} \times_3 \boldsymbol{U}_i^{(3)}. \tag{9}$$

E. Frequency Mamba Joint Modulation

As illustrated in Fig. 2(d), the FMJM module serves as the core-estimation component of the TCFM unit, where the Tucker core is inferred through joint spatial–frequency conditioning. Rather than relying on iterative tensor optimization, FMJM adopts a learnable core-estimation pipeline consisting of a Spatial SSM branch, a SERM branch, and a learnable core projection. The Spatial SSM models long-range spatial dependencies with linear complexity, whereas SERM serves as a frequency-compensation branch that provides Fourier-domain cues for attenuated frequency details. Together, they produce a spatial–frequency refined representation with enriched spatial-context and frequency-detail cues for subsequent Tucker-core estimation.

Given the precore feature $\boldsymbol{\mathcal{G}}_0$, the spatial–frequency refinement in FMJM is formulated as

$$\boldsymbol{\mathcal{Z}}_i = \alpha\boldsymbol{\mathcal{G}}_0 + \mathrm{Spatial\ SSM}\left(\boldsymbol{\mathcal{G}}_0\right) + \mathrm{SERM}\left(\boldsymbol{\mathcal{G}}_0\right) \tag{10}$$

where α is a learnable scaling factor that adaptively controls the residual contribution of the precore feature during refinement. The refined feature $\boldsymbol{\mathcal{Z}}_i$ is then mapped to the Tucker-core space through a learnable core projection, yielding the compact Tucker core $\boldsymbol{\mathcal{G}}_i$ for subsequent modewise low-rank reconstruction.

1) Spatial SSM: To capture long-range spatial dependencies with linear complexity, we introduce a Spatial SSM branch within FMJM for efficient spatial context aggregation. As illustrated in Fig. 5, the precore feature $\boldsymbol{\mathcal{G}}_0$ is first normalized to obtain $\boldsymbol{S}_0$. The SSM branch then processes $\boldsymbol{S}_0$ through a learnable linear projection ϕ_1, a 3×3 depthwise convolution (DWConv), a SiLU activation, a 2-D SSM, and layer normalization (LN), producing the spatial dependence-enhanced feature $\boldsymbol{S}_1$. Formally,

$$\begin{aligned}\boldsymbol{S}_0 &= \mathrm{LN}\left(\boldsymbol{\mathcal{G}}_0\right)\\ \boldsymbol{S}_1 &= \mathrm{LN}\left(\mathrm{SSM}\left(\mathrm{SiLU}\left(\mathrm{DWConv}\left(\phi_1\left(\boldsymbol{S}_0\right)\right)\right)\right)\right).\end{aligned} \tag{11}$$

In the modulation branch, another learnable linear projection ϕ_2 and a SiLU activation are applied to $\boldsymbol{S}_0$

$$\boldsymbol{S}_2 = \mathrm{SiLU}\left(\phi_2\left(\boldsymbol{S}_0\right)\right). \tag{12}$$

The outputs $\boldsymbol{S}_1$ and $\boldsymbol{S}_2$ are then merged through the Hadamard product, followed by a linear projection ϕ_3 to produce the output $\boldsymbol{S}_{\mathrm{out}}$

$$\boldsymbol{S}_{\mathrm{out}} = \phi_3\left(\boldsymbol{S}_1 \odot \boldsymbol{S}_2\right). \tag{13}$$

This gated fusion combines long-range spatial context with input-adaptive modulation.

2) Spectral Energy Reallocation Modulator: To compensate for the attenuation of discriminative frequency details in remote sensing imaging, SERM is employed as the frequency-aware compensation branch within FMJM. Given the precore feature $\boldsymbol{\mathcal{G}}_0$, SERM constructs two complementary Fourier-domain pathways. The radial-frequency branch adaptively recalibrates channelwise Fourier responses according to the radial-frequency profile, whereas the nonlinear mixing branch captures cross-channel dependencies in the frequency domain. The resulting responses provide frequency-compensation cues for subsequent Tucker-core modulation and are formulated as

$$\begin{aligned}\boldsymbol{F}_a &= \mathcal{F}^{-1}\left(\mathrm{RFA}\left(\boldsymbol{w}_f \odot \mathcal{F}\left(\boldsymbol{\mathcal{G}}_0\right)\right)\right)\\ \boldsymbol{F}_m &= \mathcal{F}^{-1}\left(\mathrm{PWConv}\left(\mathrm{GELU}\left(\mathrm{PWConv}\left(\mathcal{F}\left(\boldsymbol{\mathcal{G}}_0\right)\right)\right)\right)\right)\end{aligned} \tag{14}$$

where $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ denote the 2-D Fourier transform and its inverse transform, respectively, $\boldsymbol{w}_f \in \mathbb{C}^C$ is a learnable per-channel complex weight that calibrates the Fourier response of each channel, and $\odot$ denotes elementwise multiplication. GELU denotes the activation function used in the nonlinear frequency-mixing branch, while RFA denotes a bounded channelwise radial-frequency activation that recalibrates the real and imaginary Fourier responses according to the normalized radial-frequency profile.

After the inverse transformation, the two frequency-enhanced spatial responses are coupled through an interaction feature

$$\boldsymbol{F}_i = \boldsymbol{F}_a \odot \boldsymbol{F}_m. \tag{15}$$

The interaction feature captures frequency-detail components shared by the radial-frequency-enhanced and nonlinear frequency-mixing responses. The two branch responses and their interaction are then concatenated and projected by a pointwise convolution to obtain the frequency-enhanced compensation feature

$$\boldsymbol{F}_{\mathrm{out}} = \mathrm{PWConv}\left(\mathrm{Concat}\left(\boldsymbol{F}_a, \boldsymbol{F}_m, \boldsymbol{F}_i\right)\right) \tag{16}$$

where $\boldsymbol{F}_{\mathrm{out}}$ is used as the frequency-compensation term in the FMJM refinement.

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F. Kronecker-Rank Interaction Modulator

In the KRIM module, the channel-mode representation $\boldsymbol{\mathcal{X}}_i^{(3)}$ is used to generate the channel-mode factor matrix $\boldsymbol{U}_i^{(3)} \in \mathbb{R}^{C\times r_{i,3}}$ through a compact Kronecker-rank interaction. As illustrated in Fig. 2(e), KRIM avoids dense spatial flattening by first applying global average pooling to extract a channel descriptor, which is then mapped to a joint vector $\boldsymbol{w} \in \mathbb{R}^{(C+r_{i,3})K}$ through a PWConv followed by a Sigmoid activation:

$$\boldsymbol{w} = \text{Sigmoid}\left(\text{PWConv}\left(\text{GAP}\left(\boldsymbol{\mathcal{X}}_i^{(3)}\right)\right)\right) \tag{17}$$

where GAP$(\cdot)$ denotes global average pooling over the spatial dimensions.

The vector $\boldsymbol{w}$ is then split and reshaped into two compact matrices: a channel-basis matrix $\boldsymbol{W}_c \in \mathbb{R}^{C\times K}$ and a Tucker-rank basis matrix $\boldsymbol{W}_r \in \mathbb{R}^{r_{i,3}\times K}$

$$\boldsymbol{W}_c, \boldsymbol{W}_r = \text{reshape}\left(\text{split}\left(\boldsymbol{w}\right)\right) \tag{18}$$

where K denotes the Kronecker interaction rank.

The channel-mode factor matrix is generated by summing K separable channel-rank interactions

$$\boldsymbol{U}_i^{(3)} = \sum_{k=1}^{K} \boldsymbol{w}_{c,k} \otimes \boldsymbol{w}_{r,k}^T \tag{19}$$

where $\boldsymbol{w}_{c,k} \in \mathbb{R}^{C\times 1}$ and $\boldsymbol{w}_{r,k} \in \mathbb{R}^{r_{i,3}\times 1}$ are the kth column vectors of $\boldsymbol{W}_c$ and $\boldsymbol{W}_r$, respectively, with $\otimes$ denoting the Kronecker product.

The key idea of KRIM is to generate the channel-mode factor matrix through a structured interaction between channel bases and Tucker-rank bases. Each separable term models a rank-specific channel modulation pattern, while their summation adaptively combines multiple such patterns. The interaction rank K controls the tradeoff between expressiveness and compactness, avoiding dense spatial flattening while imposing a structured low-rank inductive bias. This design provides a lightweight content-adaptive factor-generation scheme for rank-aware channel modulation.

G. Multiscale Dilated Channel Attention

While conventional channel attention mechanisms, such as SE-Net [50] and ECA-Net [51], primarily rely on globally pooled descriptors to model channelwise dependencies, HSI fusion requires a better balance between spectral recalibration and the preservation of multiscale spatial textures. To this end, we design MDCA to augment BTDR by integrating heterogeneous spatial contexts with adaptive channel modulation.

Given an input feature map $\boldsymbol{F}_t \in \mathbb{R}^{H\times W\times C}$, MDCA first employs parallel depthwise dilated convolutions to capture multiscale spatial context. Using dilation rates $\mathcal{D} = \{1, 2, 3\}$, we aggregate multiscale features as

$$\hat{\boldsymbol{\mathcal{X}}} = \text{PWConv}\left(\text{Concat}_{d\in\mathcal{D}}\left(\Phi_{3\times 3}^{d}\left(\boldsymbol{F}_t\right)\right)\right) \tag{20}$$

where $\Phi_{3\times 3}^{d}$ denotes a 3×3 DWConv with dilation rate d, and PWConv fuses the concatenated features into a unified representation.

To adaptively recalibrate spectral channels, we compute a channelwise weight vector from $\hat{\boldsymbol{\mathcal{X}}}$ via global average pooling and a lightweight excitation bottleneck

$$\boldsymbol{\mathcal{X}}_w = \sigma\left(\text{PWConv}\left(\text{GELU}\left(\text{PWConv}\left(\text{GAP}\left(\hat{\boldsymbol{\mathcal{X}}}\right)\right)\right)\right)\right) \tag{21}$$

where the intermediate pointwise convolutions sequentially reduce and restore the channel dimension by a ratio ρ.

To obtain the final enhanced feature $\boldsymbol{E}_t$, MDCA modulates the multiscale features $\hat{\boldsymbol{\mathcal{X}}}$ using the derived weights $\boldsymbol{\mathcal{X}}_w$ and integrates them with the original input $\boldsymbol{F}_t$ through a residual connection

$$\boldsymbol{E}_t = \hat{\boldsymbol{\mathcal{X}}} \odot \boldsymbol{\mathcal{X}}_w + \boldsymbol{F}_t. \tag{22}$$

This design injects multiscale texture cues through dilated depthwise paths and improves spectral fidelity via channelwise recalibration.

IV. Experiments and Analysis

A. Experimental Settings

1) Datasets: We evaluate the proposed framework on five benchmark datasets, covering three simulated datasets and two real-world remote sensing fusion datasets. The simulated evaluation is conducted on three widely adopted datasets: CAVE [56], Harvard [57], and Chikusei [58]. For real-world assessment, the WHU-MHF dataset [59] is used to evaluate performance under practical HSI–MSI imaging conditions, while the WorldView-3 (WV-3) dataset [60] is used to examine generalization across different remote sensing fusion scenarios. In addition, we conduct ablation studies and stability analyses to quantify the contribution of each core component and assess the robustness of the proposed design.

The CAVE dataset consists of 32 high-quality indoor scenes captured by a generalized assorted pixel camera. Each image contains 31 spectral bands ranging from 400 to 700 nm, with a spatial resolution of 512 × 512 pixels. During model optimization, 20 images are randomly selected for training, while the remaining 12 are used for testing. The Harvard dataset comprises 77 HSIs, including 50 HSIs captured under natural daylight and 27 HSIs captured under artificial or mixed lighting. Each image contains 31 spectral bands spanning 420–720 nm, with a spatial resolution of 1392 × 1040 pixels. For consistency, a 1024 × 1024 region is cropped from the top-left corner of each image. The first 30 daylight HSIs are used for training, and the remaining 20 daylight HSIs are used for testing. The Chikusei dataset captures agricultural and urban areas in Chikusei, Japan. It consists of a single image with 128 spectral bands covering wavelengths from 363 to 1018 nm, with a spatial resolution of 2517 × 2335 pixels. A 2048 × 2048 subregion from the top-left corner is divided into 16 patches of 512 × 512 pixels, among which ten patches are used for training, and six are used for testing.

The WHU-MHF dataset serves as a specialized space-borne benchmark for HSI–MSI fusion and comprises six primary scene pairs captured by the ZY-1-02D satellite. These images feature LR-HSIs and HR-MSIs with ground sampling distances of 30 and 10 m, respectively. Following the experimental protocol in [59], five scenes are partitioned into overlapping patches of size 40 × 40 for HSIs and 120 × 120

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TABLE I

AVERAGE QUANTITATIVE RESULTS OF ALL THE METHODS ON THREE DATASETS UNDER THE ×4 AND ×8 SETTINGS. THE BEST AND SECOND-BEST VALUES ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY. THE NUMBER OF PARAMETERS (PARAMS) AND FLOPS ARE COMPUTED ON THE CAVE DATASET. "M" AND "G" DENOTE MILLIONS AND BILLIONS, RESPECTIVELY

Scale	Method	CAVE				Harvard				Chikusei				Params↓	FLOPs↓
		PSNR↑	SSIM↑	SAM↓	ERGAS↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	(M)	(G)
×4	$HySure_{15}$ [52]	43.120	0.9765	6.780	2.318	45.502	0.9817	4.102	2.839	33.218	0.8864	4.192	5.284	—	—
	$LTTR_{19}$ [53]	45.828	0.9856	5.644	1.526	46.214	0.9840	3.412	2.346	34.762	0.9142	3.748	4.920	—	—
	$GTNN_{25}$ [10]	46.917	0.9898	4.383	1.318	46.735	0.9859	3.140	1.920	37.418	0.9410	3.193	3.822	—	—
	$DHIF_{22}$ [54]	52.280	0.9973	1.840	0.746	48.270	0.9884	2.568	1.730	43.464	0.9660	2.682	2.093	22.67	56.50
	$MSST_{23}$ [15]	51.152	0.9968	2.062	0.832	48.193	0.9882	2.581	1.835	43.304	0.9654	2.707	2.196	34.39	46.45
	$DSPNet_{23}$ [35]	51.792	0.9970	2.008	0.793	48.054	0.9880	2.620	1.773	41.863	0.9574	2.732	2.396	6.06	6.60
	$MIMO_{24}$ [13]	51.492	0.9967	2.083	0.834	48.234	0.9882	2.607	1.758	42.344	0.9585	2.720	2.281	4.98	6.53
	$KNLNet_{24}$ [12]	51.816	0.9969	2.014	0.800	48.234	0.9882	2.608	1.747	43.194	0.9637	2.647	2.135	1.52	23.60
	$LTRN_{24}$ [23]	52.019	0.9971	1.950	0.769	48.307	0.9883	2.570	1.729	43.396	0.9674	2.613	2.115	0.69	**0.92**
	DCT_{24} [36]	51.482	0.9968	2.065	0.827	48.361	0.9886	2.553	1.714	43.782	0.9693	2.604	2.079	8.14	76.79
	$MCIFNet_{25}$ [43]	51.621	0.9969	2.036	0.819	48.267	0.9882	2.607	1.743	43.416	0.9656	2.689	2.096	**0.51**	2.25
	$DCINN_{24}$ [14]	52.684	0.9974	1.831	0.721	48.422	0.9886	2.546	1.711	42.494	0.9612	2.652	2.238	5.00	104.15
	$CYformer_{25}$ [55]	52.475	0.9973	1.847	0.742	48.396	0.9885	2.558	1.721	43.650	0.9701	2.652	2.238	2.10	5.82
	BFMM	**53.405**	**0.9977**	**1.664**	**0.645**	**48.610**	**0.9887**	**2.521**	**1.676**	**44.609**	**0.9714**	**2.498**	**1.939**	1.22	3.95
×8	$HySure_{15}$ [52]	40.266	0.9712	8.167	1.537	43.271	0.9778	4.652	2.006	30.612	0.8369	6.742	4.812	—	—
	$LTTR_{19}$ [53]	43.989	0.9823	5.638	1.117	44.712	0.9809	3.899	1.552	31.739	0.8998	5.486	4.358	—	—
	$GTNN_{25}$ [10]	45.062	0.9875	4.621	0.801	45.424	0.9836	3.310	1.325	32.350	0.9095	4.253	3.693	—	—
	$DHIF_{22}$ [54]	50.192	0.9961	2.119	0.484	47.414	0.9862	2.757	0.980	40.114	0.9358	3.092	1.435	22.67	56.50
	$MSST_{23}$ [15]	49.082	0.9953	2.302	0.541	47.485	0.9862	2.753	1.015	37.643	0.9243	3.522	1.916	34.39	46.45
	$DSPNet_{23}$ [35]	50.284	0.9959	2.194	0.489	47.393	0.9861	2.795	0.987	39.354	0.9318	3.083	1.540	6.06	6.60
	$MIMO_{24}$ [13]	49.043	0.9948	2.579	0.559	47.284	0.9857	2.862	1.008	38.774	0.9312	3.129	1.613	4.98	6.53
	$KNLNet_{24}$ [12]	48.974	0.9950	2.470	0.564	47.524	0.9861	2.787	0.974	40.294	0.9372	3.043	1.416	1.52	23.60
	$LTRN_{24}$ [23]	49.696	0.9954	2.344	0.522	47.527	0.9862	2.760	0.976	40.364	0.9422	2.993	1.403	0.69	**0.92**
	DCT_{24} [36]	50.503	0.9962	2.118	0.472	47.503	0.9863	2.754	0.988	40.415	0.9418	3.035	1.409	8.14	76.79
	$MCIFNet_{25}$ [43]	50.219	0.9960	2.122	0.481	47.516	0.9863	2.766	0.997	40.386	0.9421	2.996	1.411	**0.51**	2.25
	$DCINN_{24}$ [14]	50.544	0.9961	2.159	0.477	47.552	0.9863	2.758	0.970	40.164	0.9388	2.997	1.410	5.00	104.15
	$CYformer_{25}$ [55]	50.446	0.9960	2.172	0.481	47.536	0.9862	2.766	0.983	40.491	0.9423	3.012	1.398	2.10	5.82
	BFMM	**51.204**	**0.9966**	**1.980**	**0.436**	**47.686**	**0.9865**	**2.719**	**0.945**	**41.369**	**0.9498**	**2.851**	**1.282**	1.22	3.95

for multispectral images (MSIs) to facilitate model training. The remaining scene is reserved for qualitative visualization to evaluate the fusion performance in realistic remote sensing scenarios.

The WV-3 dataset is used to examine the cross-task generalization of the proposed method in the pansharpening setting. It comprises 10 000 samples derived from the PanCollection [60], with 90% used for training and 10% for validation. Each sample contains PAN/LRMS/GT triplets of sizes 64 × 64, 16 × 16 × 8, and 64 × 64 × 8, respectively. PAN images have a spatial resolution of 0.3 m, while LRMS images have a spatial resolution of 1.2 m with eight spectral bands. Testing is conducted under two settings, each with 20 samples. The reduced-resolution setting contains PAN/LRMS/GT data of sizes 256 × 256, 64 × 64 × 8, and 256 × 256 × 8, respectively, while the full-resolution setting contains PAN/LRMS data of sizes 512 × 512 and 128 × 128 × 8, respectively.

2) Data Simulation: For supervised training and evaluation on simulated datasets, we follow the standard degradation protocol in which the original HSI is treated as the ground truth (GT), while the LR-HSI and HR-MSI are synthetically generated. To obtain the LR-HSI, the GT image is blurred using a Gaussian kernel of size $p \times p$ with a standard deviation of 2, followed by spatial downsampling by a factor of p along both axes, where p denotes the spatial scale factor. The HR-MSI is generated by spectrally degrading the GT according to the spectral response function (SRF). The SRF of a Nikon D700 camera is used for the CAVE and Harvard datasets, while that of a Canon EOS 5D Mark II is applied for the Chikusei dataset. During training, image patches of size 64 × 64 are randomly extracted from the GT

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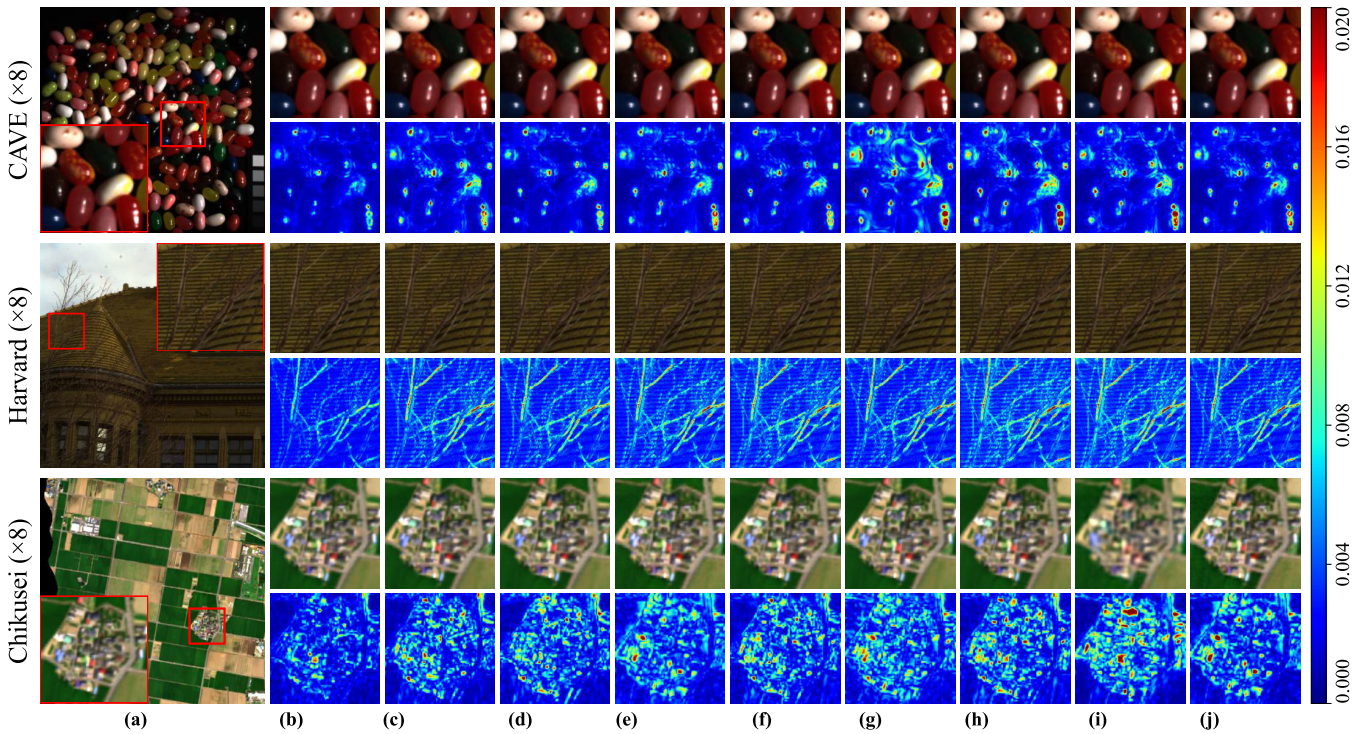


Fig. 6. Visual comparisons of selected representative methods on three (×8) datasets. The three large images present pseudocolor outputs for representative samples, with red boxes indicating regions that are selected for local magnification. For each dataset block, the top row shows enlarged pseudocolor details, and the bottom row shows the corresponding residual maps. (a) GT. (b) BFMM. (c) CYformer. (d) DCINN. (e) MCIFnet. (f) DCT. (g) LRTN. (h) KNLNet. (i) MIMO. (j) DSPNet.

HR-HSI. The corresponding LR-HSI and HR-MSI patches are subsequently generated following the above degradation process, with spatial sizes $(64/p) \times (64/p)$ and 64×64, respectively. To enrich the training samples, data augmentation techniques such as random flips and rotations are used.

3) Implementation Specifics: To ensure fairness and reliability, we follow the experimental configurations reported in the original papers where possible. All experiments were conducted within the same computational environment using the PyTorch framework on a single NVIDIA RTX 3090 GPU, thereby maintaining consistency across all compared methods. The implementation details of the proposed method are as follows. We employ the L_1 loss function in conjunction with the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The number of training epochs is set to 200, with a batch size of 16 and an initial learning rate of 3×10^{-4}.

4) Evaluation Metrics: To quantitatively assess the spatial–spectral quality of the reconstructed HR-HSI results, we adopt four standard metrics [15]: peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), spectral angle mapper (SAM), and relative dimensionless global error in synthesis (ERGAS). For the WV-3 pansharpening dataset, we use PSNR, SAM, ERGAS, and Q8 under the reduced-resolution setting, and D_s, D_λ, and QNR under the full-resolution setting. We also report the number of model parameters (Params) and floating-point operations (FLOPs) on the CAVE dataset to analyze computational efficiency.

B. Comparisons With State-of-the-Art Methods

In this section, we conduct a comprehensive comparison with representative model-based methods and recent DL-based methods to evaluate the proposed BFMM. The model-based baselines include HySure [52], LTTR [53], and GTNN [10]. The DL-based baselines include DHIF [54], KNLNet [12], MSST [15], DSPNet [35], MIMO [13], LTRN [23], DCT [36], MCIFNet [43], DCINN [14], and CYformer [55]. The evaluation covers quantitative accuracy, visual quality, and computational efficiency, providing an overall assessment of BFMM.

1) Quantitative Comparison: Table I presents the quantitative results of the proposed method and 13 competing approaches. The experimental results show that BFMM achieves the best overall results among all compared methods across the three standard datasets. Under the ×4 scaling factor, our approach yields PSNR gains of 0.721, 0.188, and 0.827 dB on the CAVE, Harvard, and Chikusei datasets, respectively, compared with the second-best method. These quantitative improvements validate the effectiveness of BFMM in increasing reconstruction fidelity and reducing reconstruction errors. Moreover, with respect to the SAM metric, our method achieves reductions of 0.167, 0.025, and 0.106 on the same datasets, respectively. Similar advantages are observed under the ×8 scaling factor, where BFMM achieves PSNR gains of 0.660, 0.134, and 0.878 dB, together with SAM reductions of 0.138, 0.034, and 0.142 on the three datasets. These results indicate that BFMM effectively preserves spectral characteristics and mitigates spectral

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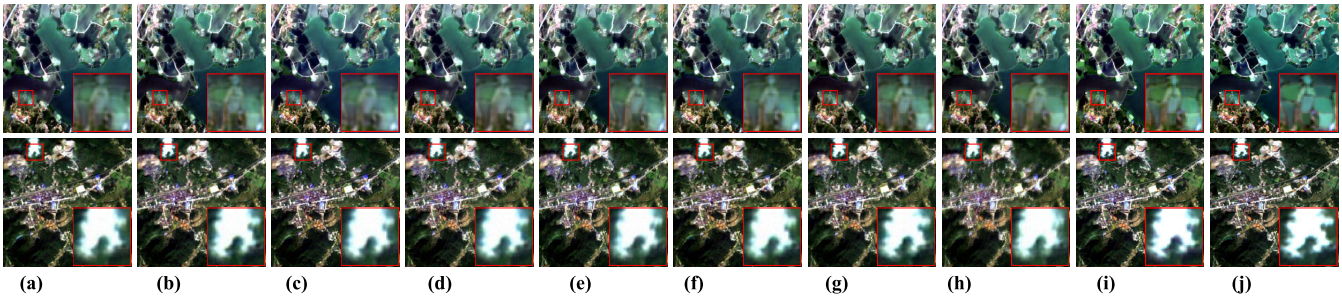


Fig. 7. Visual comparisons on the real-world WHU-MHF dataset. The figure displays representative results for a lakeside scene in the first row and an urban scene in the second row. The large images depict the pseudocolor fusion outputs, with red rectangles highlighting regions of local magnification. (a) DSPNet. (b) MIMO. (c) KNLNet. (d) LRTN. (e) DCT. (f) MCIFnet. (g) DCINN. (h) CYformer. (i) BFMM. (j) HRMS.

TABLE II

AVERAGE QUANTITATIVE RESULTS ON THE WV-3 DATASET (20 REDUCED-RESOLUTION AND 20 FULL-RESOLUTION SAMPLES). THE BEST AND SECOND-BEST ENTRIES ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	Reduced-Resolution Metrics				Full-Resolution Metrics		
	PSNR↑	SAM↓	ERGAS↓	Q8↑	$D_\lambda \downarrow$	$D_s \downarrow$	QNR↑
LightNet_{22} [29]	36.235 ± 1.608	3.311 ± 0.613	2.524 ± 0.618	0.898 ± 0.086	0.0647 ± 0.0316	0.0367 ± 0.0039	0.9009 ± 0.0275
LAGConv_{22} [61]	36.790 ± 1.597	3.247 ± 0.616	2.366 ± 0.527	0.896 ± 0.088	0.0503 ± 0.0339	0.0371 ± 0.0038	0.9144 ± 0.0302
CMLNet_{23} [62]	37.009 ± 1.623	3.126 ± 0.549	2.289 ± 0.510	0.898 ± 0.091	0.0458 ± 0.0323	0.0344 ± 0.0037	0.9213 ± 0.0288
LGTEUN_{23} [63]	37.158 ± 1.839	3.083 ± 0.517	2.210 ± 0.446	0.905 ± 0.088	0.0380 ± 0.0277	0.0367 ± 0.0045	0.9266 ± 0.0233
BiMPan_{23} [64]	36.443 ± 1.693	3.233 ± 0.573	2.441 ± 0.577	0.898 ± 0.090	0.0383 ± 0.0143	0.0370 ± 0.0046	0.9260 ± 0.0119
DISPNet_{24} [65]	37.430 ± 1.762	2.899 ± 0.518	2.175 ± 0.446	0.910 ± 0.085	0.0481 ± 0.0296	0.0381 ± 0.0045	0.9155 ± 0.0253
MSAN_{25} [66]	37.203 ± 1.800	3.077 ± 0.550	2.236 ± 0.493	0.900 ± 0.089	0.0420 ± 0.0283	0.0357 ± 0.0038	0.9238 ± 0.0250
ARNet_{25} [67]	37.575 ± 1.724	2.909 ± 0.534	2.134 ± 0.470	0.911 ± 0.084	0.0468 ± 0.0298	**0.0330 ± 0.0042**	0.9216 ± 0.0255
BFMM	**37.946 ± 1.786**	**2.761 ± 0.496**	**2.102 ± 0.447**	**0.914 ± 0.083**	**0.0317 ± 0.0176**	0.0335 ± 0.0043	**0.9374 ± 0.0128**

distortion during reconstruction. Overall, BFMM achieves competitive performance across the evaluation metrics while maintaining a favorable accuracy–complexity tradeoff compared with most high-accuracy baselines. This demonstrates the effectiveness of integrating deep networks with low-rank tensor modeling.

2) Qualitative Comparison: Fig. 6 shows a qualitative comparison of BFMM with selected representative methods across the CAVE, Harvard, and Chikusei datasets under the $\times 8$ setting. On the CAVE dataset, BFMM effectively recovers fine textures and small-scale structures, such as the surface details of the jelly beans. The texture and boundaries of the jelly beans are faithfully preserved, with accurate colors and sharp edges. The residual map indicates that the residual error of BFMM is more sparsely and locally distributed than that of the other methods, which exhibit larger errors over broader regions. Compared with the other methods, BFMM introduces fewer artifacts and less noise, while retaining more details and better visual clarity. On the Harvard dataset, BFMM demonstrates clear advantages in reconstructing intricate tree-branch textures embedded in complex background structures. Competing methods often fail to preserve the connectivity and sharpness of fine twigs, resulting in visible blurring or fragmentation artifacts. In contrast, BFMM maintains clearer structural integrity and sharper edges. This visual assessment is corroborated by the residual maps, where BFMM exhibits lower reconstruction error magnitude, effectively suppressing spatial distortion and achieving higher spatial–spectral fidelity. For the Chikusei dataset, BFMM accurately restores the delineation between land and buildings, particularly in the transition regions between urban and agricultural areas, thereby reducing image blurring and oversmoothing. The residual map shows that the errors generated by BFMM are sparse and localized, indicating the robustness of the method in large-scale scene reconstruction, especially in distinguishing different land-cover categories, such as buildings and fields. Overall, BFMM maintains fine texture details and introduces fewer artifacts and less noise across the datasets, especially in high-contrast scenes and complex backgrounds.

Furthermore, to evaluate the generalization performance of BFMM in complex real-world scenarios, Fig. 7 presents visual comparisons on the WHU-MHF dataset. In the lakeside scene in the first row, BFMM precisely delineates the intricate outlines of the fields surrounding the lake, effectively suppressing the edge blurring observed in other methods. In the urban scene in the second row, the proposed method reconstructs the fine textures and distinct morphological outlines of clouds more effectively, whereas other methods suffer from noticeable oversmoothing and loss of structural information. Overall, while improving spatial resolution, BFMM maintains favorable shape fidelity and spectral consistency, further supporting its applicability to real-world fusion scenarios.

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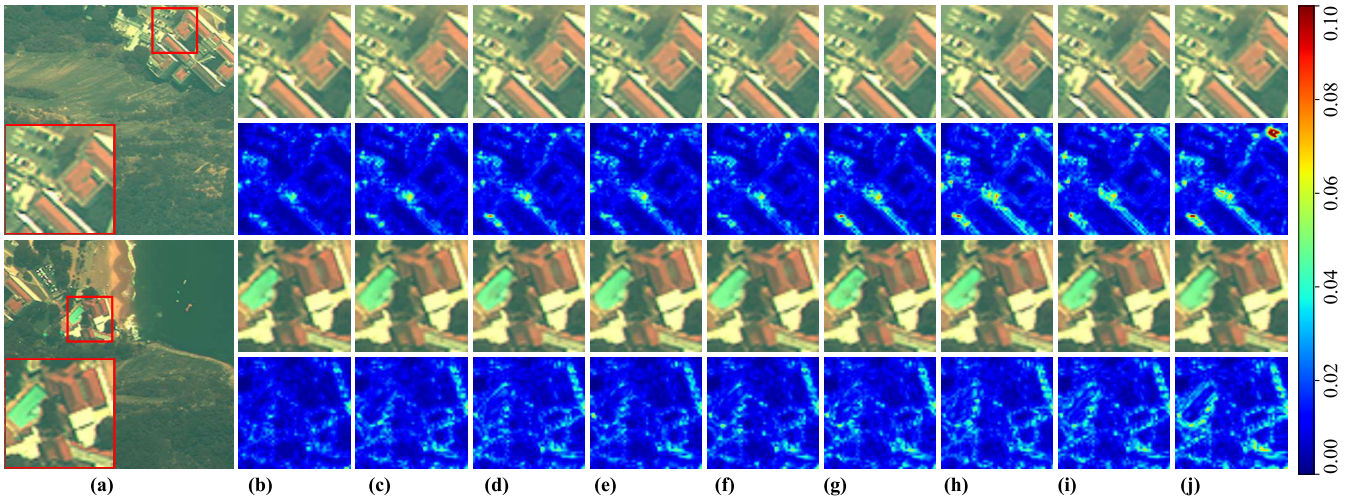


Fig. 8. Visual comparisons of all compared methods for the pansharpening task on the WV-3 dataset under the reduced-resolution setting. (a) GT. (b) BFMM. (c) ARNet. (d) MSAN. (e) DISPNet. (f) BiMPan. (g) LGTEUN. (h) CMLNet. (i) LAGConv. (j) LightNet.

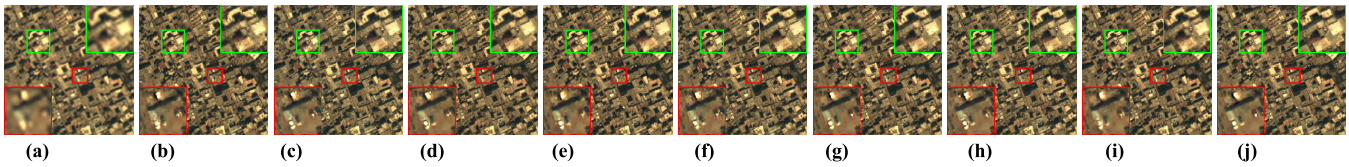


Fig. 9. Visual comparisons of the LRMS input and all compared methods on the WV-3 dataset under the full-resolution setting, with two regions enlarged for detailed observation. (a) LRMS. (b) LightNet. (c) LAGConv. (d) CMLNet. (e) LGTEUN. (f) BiMPan. (g) DISPNet. (h) MSAN. (i) ARNet. (j) BFMM.

C. Comparison on the Pansharpening Dataset

In this section, we further evaluate the proposed method on the WV-3 pansharpening dataset to examine its cross-task generalization. The BFMM results on WV-3 are reported in Table II. BFMM achieves the best results on most metrics, indicating a favorable balance between spatial enhancement and spectral preservation. Under the reduced-resolution setting, it obtains the best performance on PSNR, SAM, ERGAS, and Q8. Under the full-resolution setting, compared with the second-best value for each metric, BFMM reduces D_λ by 0.0063 and increases QNR by 0.0108, demonstrating its effectiveness and robustness in the real full-resolution pansharpening scenario.

To provide an intuitive assessment, Fig. 8 presents visual comparisons on reduced-resolution test samples. The fusion results produced by BFMM exhibit sharper details, while both spectral distortion and spatial artifacts are effectively suppressed. Visual results on full-resolution samples are shown in Fig. 9, highlighting the ability of BFMM to inject fine spatial information from the high-resolution PAN input while preserving the natural appearance and spectral integrity of the targets. In summary, the quantitative and qualitative evaluations demonstrate that BFMM achieves favorable performance on the WV-3 dataset, confirming its reconstruction capability and cross-task generalization.

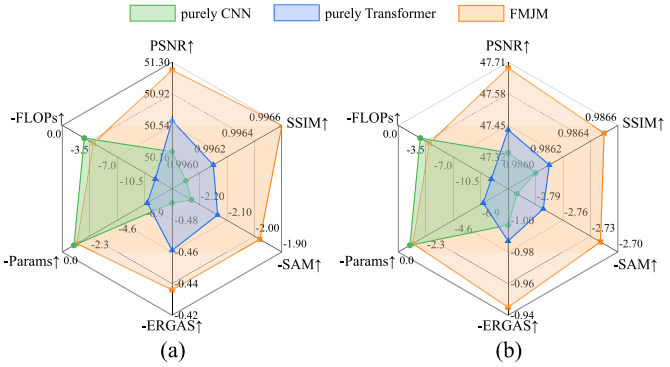


Fig. 10. Ablation study on FMJM. (a) and (b) Experimental comparisons on the CAVE (×8) and harvard (×8) datasets, respectively.

TABLE III
ABLATION OF BFMM KEY COMPONENTS ON THE CAVE (×8) DATASET. THE BEST AND SECOND-BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Baseline	FMJM	KRIM	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Params (M)↓	FLOPs (G)↓
✓			50.121	0.9959	2.248	0.497	**0.81**	**2.65**
✓	✓		<u>50.891</u>	<u>0.9963</u>	<u>2.050</u>	<u>0.456</u>	1.12	3.89
✓		✓	50.598	0.9961	2.134	0.461	<u>0.91</u>	<u>2.68</u>
✓	✓	✓	**51.204**	**0.9966**	**1.980**	**0.436**	1.22	3.95

D. Ablation Study

1) Ablation of Crucial Components: To evaluate the contributions of the proposed modules, we conduct ablation experiments on BFMM variants. The results on the CAVE dataset are reported in Table III. To ensure fairness, when a module is removed, it is replaced with a baseline. With the integration of FMJM, the PSNR increases from 50.121

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TABLE IV

ABLATION STUDY OF TENSOR DECOMPOSITION SCHEMES ON THE CAVE (×8) DATASET. THE BEST AND SECOND-BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

FMJM	CP	BTD	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Params (M)↓	FLOPs (G)↓
✓			<u>50.606</u>	<u>0.9962</u>	<u>2.170</u>	<u>0.450</u>	**1.07**	**3.73**
✓	✓		50.234	0.9961	2.203	0.486	<u>1.12</u>	<u>3.74</u>
✓		✓	**51.204**	**0.9966**	**1.980**	**0.436**	1.22	3.95

TABLE V

ABLATION STUDY OF DIFFERENT ATTENTION MECHANISMS ON THE CAVE (×8) DATASET. THE BEST AND SECOND-BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Baseline	SE	ECA	MDCA	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Params (M)↓	FLOPs (G)↓
✓	✓			<u>51.031</u>	<u>0.9965</u>	<u>2.036</u>	<u>0.441</u>	<u>1.20</u>	<u>3.77</u>
✓		✓		50.975	0.9964	2.042	0.443	**1.19**	**3.76**
✓			✓	**51.204**	**0.9966**	**1.980**	**0.436**	1.22	3.95

to 50.891, the SSIM improves slightly, and both SAM and ERGAS decrease. Fig. 10 presents radar charts that further illustrate the performance differences caused by replacing FMJM with purely CNN or Transformer branches. Similarly, adding KRIM alone also yields steady gains. Combining both modules delivers the best overall performance, with the PSNR reaching 51.204 and the ERGAS decreasing to 0.436. These findings indicate that the frequency-aware long-range modeling of FMJM and the channel-rank interaction of KRIM are complementary.

2) Comparative Study of Tensor Decomposition Schemes: To evaluate whether BTD preserves the low-rank prior while maintaining sufficient intrablock interactions, we compare it with CP decomposition within the same FMJM-based architecture. As shown in Table IV, the CP-based variant has slightly lower parameter and FLOP counts than the BTD-based variant, but it yields inferior reconstruction performance across all accuracy metrics. This suggests that the strictly rank-1 separable structure of CP decomposition is overly restrictive for modeling high-order spatial–spectral dependencies in HSI fusion. In contrast, BTD represents the tensor through a few low-rank block terms, each using a compact core and factor matrices. This approach achieves significantly higher reconstruction quality, reaching a PSNR of 51.204 dB alongside consistent improvements in SAM and ERGAS. These results demonstrate that the BTD core provides sufficient flexibility to capture higher-order mode couplings, thereby improving the expressive capacity of low-rank tensor modeling while maintaining reconstruction fidelity.

3) Effectiveness of the Proposed MDCA: To further examine the contribution of MDCA, we compare it with two widely used channel-attention modules, SE [50] and ECA [51]. Table V shows that MDCA consistently improves reconstruction accuracy over both SE and ECA. MDCA achieves 51.204-dB PSNR, exceeding SE and ECA by 0.173 and 0.229 dB, respectively. This improvement requires a modest parameter increase of 0.03M over ECA, suggesting that the added multiscale spatial aggregation provides a favorable accuracy–complexity tradeoff for HSI–MSI fusion.

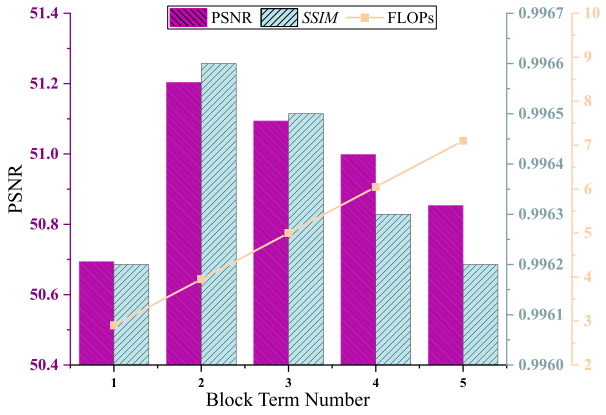


Fig. 11. Effect of the block term number L on reconstruction accuracy and computational complexity on the CAVE (×8) dataset. Bars denote PSNR and SSIM, and the line denotes FLOPs as L varies from 1 to 5. The left y-axis corresponds to PSNR, while the right y-axes correspond to SSIM and FLOPs, respectively.

TABLE VI

EVALUATION OF THE BLOCK TERM NUMBER L ON THE CAVE (×8) DATASET. THE OPTIMAL VALUES ARE INDICATED IN BOLD, AND THE SECOND-BEST VALUES ARE UNDERLINED

Setting	Set	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Params (M)↓	FLOPs (G)↓
block term number L	1	50.694	0.9962	2.119	0.462	**0.92**	**2.90**
	2	**51.204**	**0.9966**	**1.980**	**0.436**	<u>1.22</u>	<u>3.95</u>
	3	<u>51.094</u>	<u>0.9965</u>	<u>2.022</u>	<u>0.442</u>	1.52	5.00
	4	50.999	0.9963	2.045	0.452	1.83	6.05
	5	50.854	0.9962	2.070	0.457	2.13	7.10

4) Hyperparameter Settings: To investigate the effect of the block term number L on performance, we fix the other hyperparameters and training protocol and evaluate $L \in \{1, 2, 3, 4, 5\}$ on the CAVE (×8) dataset. As shown in Table VI, increasing L from 1 to 2 significantly improves all accuracy metrics. For example, the PSNR increases from 50.694 to 51.204, and the SAM decreases from 2.119 to 1.980, indicating that an overly small block-term number limits the representational capacity. However, when $L \geq 3$, the performance consistently declines across the metrics, with the most pronounced degradation observed in SAM and ERGAS. Meanwhile, both the parameter counts and FLOPs increase monotonically with L, introducing unnecessary computational overhead without performance gains. Considering both accuracy and computational efficiency, we set $L = 2$ in all reported experiments. In addition, the trend curves of key metrics with respect to L shown in Fig. 11 provide a more intuitive confirmation of this choice.

V. CONCLUSION

In this work, we introduce BFMM, a low-rank tensor modulation framework for HSI fusion, motivated by two practical issues: the frequency insensitivity of long-range modeling and the underutilization of low-rank spatial–spectral structures in hyperspectral features. BFMM represents features as sums of Tucker components under BTD and couples frequency-aware state-space modulation with Kronecker-guided factor estimation to keep representations compact and interpretable.

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Across three public datasets, BFMM achieves state-of-the-art reconstruction fidelity with a favorable accuracy–complexity tradeoff. Experiments on the real-world WHU-MHF dataset further validate its applicability under practical imaging conditions, while the WV-3 pansharpening results demonstrate its cross-task generalization. Overall, BFMM offers an efficient and structurally interpretable solution for HSI fusion.

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